

AEP-X — Microservices Architecture v2 Addendum

Knowledge Service, Verification Engine, Cost Optimiser, ML Integration — plus the canonical L0–L5 cache

Source [AEP-X Ultra \(1\).docx](#) Extends [Microservices-Implementation-Guide.html](#)

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1. Reconciliation — What This Addendum Adds and Why

The original Microservices-Implementation-Guide reconciled a short "Volume 9" outline into 7 services plus a backlogged Marketplace Engine. This new source, read in full, adds four engine families that outline never named at build-detail level:

New engine family	Source	Already covered?
Knowledge Service	Sprint 2.3 MVP; RFC-0005 "Knowledge"	No — Memory Service only covers session memory
ML Integration Layer (Prediction/Learning/Optimisation)	Early-pass "ML Integration Layer": Classification/Prediction/Recommendation/Continuous Learning	No — ADLC Plan §12 mentions it only as an operations concept
Anti-Hallucination Engine V2 / Verification Engine	Sprint 4.3: Claim Extraction → Evidence Correlation → Truth Scoring → Confidence → Citation	Partially — Instructional Manual §4.4 conflated it with Safety Engine's response contract
Universal Cost Optimiser	Utility Score = Accuracy+Trust+Compliance+Performance – Cost–Latency–Energy–Risk	No — cache-first exists as a principle, no service computes it at request time

Scope discipline check: these are new Tier 2 surface, same tier as Workflow/Safety/Governance. Build order matters more than completeness — ordered below by dependency and by how directly they serve the platform's core economic claim (80% LLM-cost reduction, ADLC Plan §9), not by how the source happens to list them.

ARCHITECTURE

2. Updated Architecture Overview

Tier 1 unchanged · Tier 2 extended with 4 services · Tier 3 gains pgvector

- **Tier 1 (unchanged):** Agent Registry, Trust Authority, Discovery Service, Memory Service.
- **Tier 2 (extended):** Workflow Engine, Safety Engine, Governance Engine, Marketplace Engine (backlog) + **Knowledge Service, Cost Optimiser Service, ML Integration Service, Verification Engine.**
- **Tier 3 (extended):** PostgreSQL, Redis, Neo4j, Kafka + **pgvector** on the existing PostgreSQL instance.

The dependency rule holds: no Tier 2 service calls another Tier 2 service synchronously — the new services communicate via Kafka events, same as Workflow/Safety/Governance already do.

3. Service Catalogue Additions

Service	Owns	Exposes	Depends on	Data store
Knowledge Service	Validated facts, embeddings	POST/GET /knowledge , /knowledge/search	Memory Service (read); Neo4j	PostgreSQL + pgvector
Verification Engine	Evidence correlation, truth/confidence scores	POST /verify	Knowledge Service; consumes workflow.completed	PostgreSQL
Cost Optimiser Service	Utility Score computations	POST /route/decide	Trust Authority, Safety Engine	Redis (decision cache only)
ML Integration Service	Prediction/recommendation models	POST /predict , /recommend	Consumes workflow.completed , safety.flagged , verification.completed	PostgreSQL + object storage

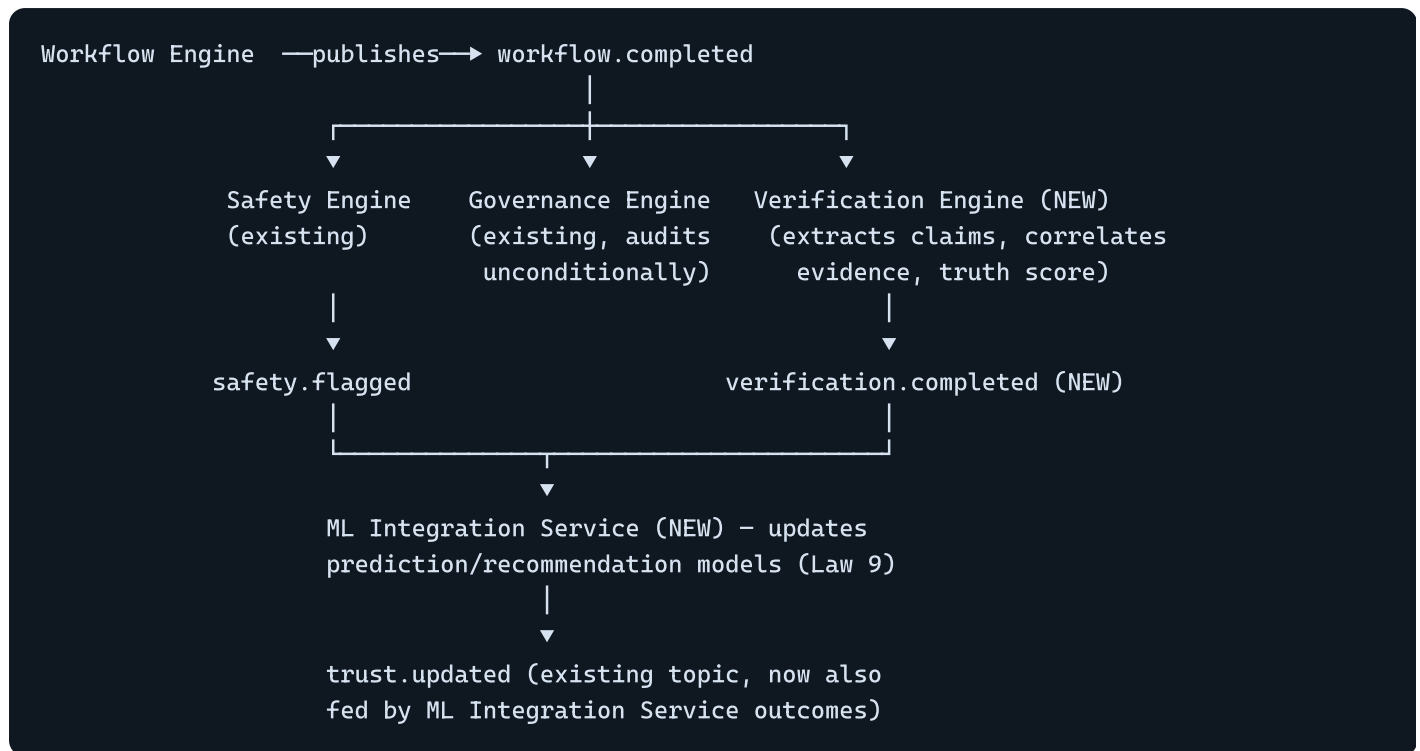
The one rule that still applies: every row owns its own schema. Cost Optimiser Service needs a trust score on every request — call the Trust Authority API and cache the response; never read `trust_scores` directly.

4. Communication Patterns — Additions

4.1 Cost Optimiser — the one synchronous exception in Tier 2

Routing decisions must be available *before* an LLM call is made, so `POST /route/decide` is synchronous from the Gateway, budgeted at <20 ms. This is the only synchronous Tier 2 call in the system — if a second one gets proposed later, treat that as a signal to re-examine the tier boundary, not a precedent.

4.2 The extended event flow



4.3 Cache/memory layer implementation, now canonical

Instructional Manual §3.5 built only **L1 Session Cache, TTL = 1 hour** as an explicit reduced-scope MVP. This addendum extends to the full model resolved in ADLC Plan §15.3, TTLs pinned by this source's Sprint 2.2 spec:

Layer	Scope	TTL
L0 Prompt Cache	Exact-match prompt+context hash	5 minutes
L1 Session Cache	Per-session (already built)	1 hour
L2 Agent Cache	Per-agent, cross-session	24 hours
L3 Workflow Cache	Per-workflow-definition reuse	7 days
L4 Organisation Cache	Cross-agent, per-organisation	30 days
L5 Federation Cache	Cross-organisation	Policy-controlled (Governance Engine decides per call)

5. Event Contracts — Additions

Topic	Producer	Consumers	Payload
<code>verification.completed</code>	Verification Engine	Governance Engine, ML Integration Service	<code>{workflow_id, claim_count, truth_score, confidence_band, citations[]}</code>
<code>ml.prediction.made</code>	ML Integration Service	Cost Optimiser Service (optional)	<code>{prediction_type, target_id, value, confidence, model_version}</code>
<code>knowledge.updated</code>	Knowledge Service	Verification Engine	<code>{knowledge_id, source, updated_fields[], ts}</code>

6. Confidence Bands — Verification Engine

Reuses the exact band structure the source repeats consistently across every pass — one of the few things that didn't drift between drafts:

```
CONFIDENCE_BANDS = [(95, "GREEN"), (80, "AMBER"), (60, "RED"), (0, "GREY")]
# GREEN=Verified, AMBER=Inference, RED=Challenge, GREY=Risk
```

This feeds the same `{answer, confidence, evidence[], verification_status}` contract Safety Engine already returns — a deeper pipeline with a real truth score, not a competing contract.

7. Definition of Done — Microservices v2 Milestone

- Knowledge Service, Verification Engine, Cost Optimiser Service, and ML Integration Service all report `"status": "ok"` on `/health`.
- Cache Service serves all six layers (L0–L5) with the TTLs above; an L5 policy-denied request returns a clear reason, not a silent fallback.
- `POST /route/decide` responds in <20 ms p95 with an explainable Utility Score breakdown.
- A workflow completion produces a real `verification.completed` event, consumed by ML Integration Service within 1 second.
- Recorded in `governance/decisions/`, noting Marketplace Engine and connector layers remain backlog.